

C Programming Interview Questions

Memory :

How the malloc internally work? How do you implement malloc ?

How to know size of memory allocated by malloc using pointer?

Why the first two parameters are of type void in memcpy?

When an unallocated pointer is dereferenced ,who will check that this is invalid instruction.

how memory leak is detected and various ways of handling

Where do const variables get stored? (literal segment)

In a Process Address Space what if stack or heap collide?

Will there be any compilation errors while trying to print the below?will there be any warning if yes when at compile time or run time?

```
int *ptr=NULL;
printf("%d",*ptr);
```

In processes address space where is string "MY_STRING" stored ?

Void main()

```
{
    char *str_ptr="MY_STRING";
    printf("%s",str_ptr);
}
```

Tell me the logic how can you find depth of a stack?

Structure:

Is it possible to declare a function in structure? How we can declare if we want?

Are #pragmas Generic or processor specific

Struct test { int a; int b; }; Assume you don't know the contents of structure, and you have the base address of test structure alone, how to do access the data inside it ??

Volatile:

About volatile keyword,how volatile ensure that to read current updated value.

What is volatile keyword? And where we can use?

Does synchronization techniques are required for Volatile variable?

Others:

Differentiate between static and extern storage classes. Can they be used together?

Can a static variable be declared inside a header file? What are the implications? >> Each .c file including the header will get its own copy of the static variable.This can lead to code bloat and hard-to-debug behavior.

Why should you never use register for memory-mapped I/O?

Explain cinit and bss section

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Difference between #define and CONST, which one will you chose in programming?

How can you make a header file not to include multiple times?

WAP to know how stack grows in upward or downward directions?

WAP to convert ACII to Interger?

Why ASCII number is required?

Given Two borads and with exactly same CPU's should they require different compilers?

What is the problem in following code

```
char * p = 'a';
```

How to fix it?

Programs:

1. How to find the given number is power of 2

2. Set a bit / Clear a bit / Toggle a bit?

3. How to find the given linked list is palindrome or not? (logic is required)

4. How many loops are required to find the given linked list is Palindrome or not?

5. Write a program to find the number of bit's set in a given number?

6. Write a program for set clear toggle a bit and find the given bit is set or not?

7. Write a macro to swap the nibbles in the word ?

Ex:0x1234 output:2143

8. Need the out put to be printed as 10.

```
int a=10;
```

```
main()
```

```
{
```

```
    int a=20;
```

```
    printf("%d",a);
```

```
}
```

what is the output.

9. main()

```
{
```

```
    int *p= &a;
```

```
    int a= 5;
```

```
    printf("%d %d", *p,a);
```

```
}
```

what is the output.

10. main()

```
{
```

```
    int a= 5;
```

```
    int *p= &a;
```

```
    printf("%d %d", *p,a);
```

```
}
```

11. main()

```
{
```

```
    int a= 5;
```

C Programming Interview Questions

```
        int *p= &a;
        *p++;
        p--;
        printf("%d %d", *p, a);
    }
```

```
12. main()
{
    int a= 5;
    int *p= &a;
    (*p)++;
    p--;
    printf("%d %d", *p, a);
}
```

13. const volatile int *ptr;
*const volatile int ptr;
What is the diff ?.

14. which pointer value we can modify.
const volatile int *ptr;
*const volatile int ptr;

15. Find a word from string

16. How to find if there is in repeating node in linked list
17. For a given array gather all 1's to one side and 0's to one side.
18. In a given byte interchange pair of bits
19. Reverse a word in a string
20. Write program to Insert element in static array.

```
    Ex: int a[20] = {1,2,3,5},
        Given input element is 4,
        answer is {1,2,3,4,5}
```

21. Write a function in library file that function should return a 32bit
22. Write a program to find how many bit to toggle in 2 binary numbers so that they become equal

a. program for to get the number of bits toggle in 2 binary numbers and toggle them to make the numbers equal

23. write a C programme to set bits from 3 to 7 positions to 1 in a 32-bit given value

Condition is at a time only one bit from 3 to 7 bit positions to be 1 and if not all bits to be reset.?

```
// file1.c
#include <stdio.h>
```

```
static int counter = 0;
```

```
void increment() {
    counter++;
}
```

```
// file2.c
```

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```
#include <stdio.h>
```

```
extern int counter;
```

```
void show() {  
    printf("Counter = %d\n", counter);  
}
```

Problem: counter is declared static in file1.c - it has internal linkage, so file2.c cannot access it using extern.

✓ Fix: Remove static if it needs to be shared, or redesign logic to access it via functions only.

Ans. string "MY_STRING" stored at read only Text segment and str_ptr stored at data segment.

16. I

17.

18. Create a pointer pointing to integer type point that to char array containing 10 elements?

19. The above created pointer if incremented ++ which element of the array it points to?

20. Write an instruction such that integer pointing should point to first element of the array on increment?

21.

22.

23. Determine the minimum stack size required for given prog by end of main function?

```
Void main()  
{  
    int x,y;  
    char c;  
    fun1();//Assuming forward declaration is done.  
    fun2();//Assuming forward declaration is done  
    fun3();//Assuming forward declaration is done  
}
```

```
void fun1()  
{  
    int a,b,c;  
    c=a+b;  
}
```

```
void fun2()  
{  
    char c;  
}
```

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```
void fun3()  
{  
}
```

24. what is the internal implementation/mechanism made to work volatile to get the updated value?

25. Difference between #define and CONST, which one will you chose in programming?

26. Tell me the logic how can you find depth of a stack?

Example In a program their is fool() function calling foo2() function calling foo3() in any function

write a logic to know the depth of stack.

27. How C programme memory address space is allocated?

(stack,heap ,data segment,code segment

28. How can you make a header file not to include multiple times?

29. WAP to know how stack grows in upward or downward directions?

30. WAP to convert ACII to Interger?

31. Why ASCII number is required?

32. Given Two borads and with exactly same CPU's should they require different compilers?

33. how this programme looks after preprocessing?

```
#include<stdio.h>
```

```
void main ()
```

```
{
```

```
int a;
```

```
}
```

34. The project done by you are in linux platform if gcc is the compiler used what is the preprocessor used?

35. what is the job of preprocessor?

36. how variable 'a' get allocated memory in assembly level language?

```
main()
```

```
{
```

```
int a;
```

```
}
```

37. In printf what it the purpose of %X ?

38. What is the size of integer variable on 32bit and 64bit machine?

39. Structure

```
{
```

```
Int a,
```

```
Char c;
```

```
Int b;
```

```
}
```

What is the size of the structure.

40. If a global variable defined as int V1=100 in programm_1 and modified this V1=200 in programme_2.what will be printed in Programme_1 and Programme_2.

41.char *name="WIPRO";

sizeof(*name) = ? (Ans :1)

sizeof("WIPRO")= ? (Ans :6)

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- sizeof(name)= ? (Ans :4)
42. Declare a function pointer which takes 2 arguments as char pointers, and returns an integer pointer?
43. For above function pointer assign the function to be called?

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Set a bit / Clear a bit / Toggle a bit?

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How many loops are required to find the given linked list is Palindrome or not?

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```
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```
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```

```
    int a= 5;
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```
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```

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}
```

what is the output.

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main()
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main()
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```
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```
    int a= 5;
```

```
    int *p= &a;
```

```
    *p++;
```

```
    p--;
```

```
    printf("%d %d",*p,a);
```

```
}
```

```
main()
```

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    int a= 5;
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```
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Reverse a word in a string

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